

ML/DL Project: Salient Object Detection (SOD)

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- Salient Object Detection (SOD) identifies the most visually important objects in an image based on human attention
- It is a fundamental task in Computer Vision, going beyond simple classification
- The goal is to build a complete end-to-end Deep Learning pipeline from scratch
- Input: image $\rightarrow$ Output: saliency mask (pixel-level segmentation)
- Applications:**
 - Image cropping and automatic editing
 - Medical imaging and diagnostics
 - Robotics and autonomous systems
 - Visual search and content retrieval

Dataset & Data Preparation

Dataset: DUTS

- ~10,500 training images
- Pixel-accurate saliency masks (ground truth labels)
- One of the most widely used datasets for SOD tasks

Data Preprocessing:

- Resize all images to 128×128 pixels
- Normalize pixel values to range [0, 1]
- Convert images to RGB format
- Convert masks to grayscale for binary segmentation
- Implemented using PyTorch Dataset and DataLoader

Data Splitting:

- 70% Training set
- 15% Validation set
- 15% Test set

Model Architecture

Baseline Model

- Simple CNN encoder–decoder architecture
- 2 encoder convolutional layers
- No Batch Normalization or Dropout
- Limited feature extraction capability
- Used as a performance baseline

Improved Model

- Deeper 3-layer encoder architecture
- Batch Normalization for stable training
- Dropout (0.3) to reduce overfitting
- Enhanced feature learning capability
- Symmetric decoder for better reconstruction

Loss Function & Optimizer

- BCEWithLogitsLoss (binary pixel-wise segmentation loss)
- Adam optimizer (learning rate = $1e-4$)
- Stable and efficient optimization for deep learning models

Training Process

Training Process

- Forward pass → Loss computation → Backpropagation → Optimizer update
- Training performed in mini-batches using DataLoader
 - Two phases:
 - Train mode (model learning)
 - Validation mode (performance evaluation without gradients)

Training Configuration

- Baseline model: 5 epochs
- Improved model: 20 epochs
- Optimizer: Adam ($lr = 1e-4$)
- Loss function: BCEWithLogitsLoss

Evaluation Metrics

- IoU → overlap between predicted and ground truth masks
- Precision → accuracy of predicted salient pixels
- Recall → how well real objects are captured
- F1 Score → balance between Precision and Recall

Results & Final Conclusion

Results

- Baseline Model:
 - IoU: 0.2882
 - F1 Score: 0.4425
- Improved Model:
 - IoU: 0.4547
 - F1 Score: 0.6231

Analysis

- Significant improvement across all metrics
- +57.8% increase in IoU
- More accurate and cleaner segmentation masks
- Improved model generalizes better on unseen data

Conclusion

- The project successfully implements a full SOD pipeline from scratch
- Batch Normalization, Dropout, and deeper architecture improve performance
- The system works effectively for real-time inference
- Provides a strong foundation for future improvements (U-Net, data augmentation)